2081:

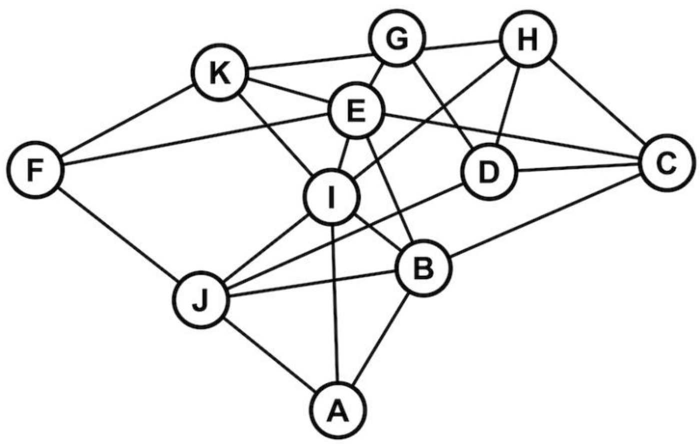
Long Answer Questions: [10 marks each]

1. How do you define optimal solution? Does greedy algorithm always guarantee optimal solution? Given the string "SUPER DUPER CSIT", use a Greedy algorithm to build a Huffman tree.
2. What is order statistics? Write and analyze the algorithm for randomized quick sort.
3. Distinguish between dynamic programming and memorization. Parenthesize the matrices A(30 × 1), B(1 × 40), C(40 × 10) and A(10 × 15), for computing matrix multiplication using dynamic programming.

Short Answer Questions: [5 marks each]

4. Solve the recurrence relation $T(n) = 2T(n/2) + n$ using recursion tree method.
5. Find the best and worst case for Bubble sort.
6. Using Extended Euclidean Algorithm, find the GCD of 12 and 16.
7. Find all possible subsets of the integers that sum to 21 in the array {5, 6, 10, 11, 15} using back tracking technique.
8. Define class P and NP problem. Why do we need approximation algorithms? Justify.
9. State the time and space complexity for sequential search. Write the rules for master theorem for finding asymptotic bounds.

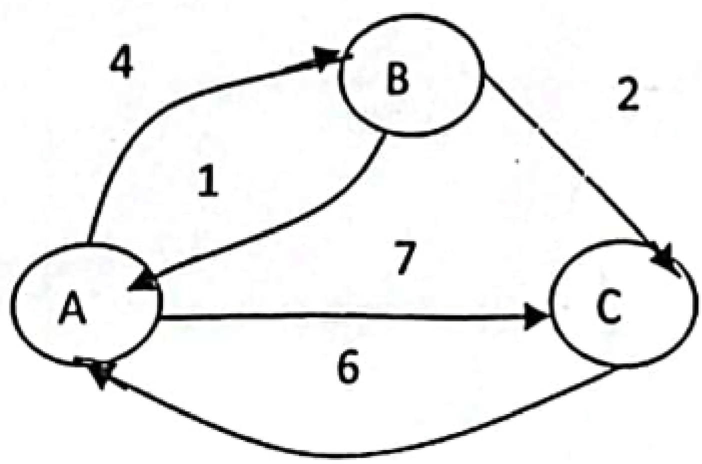
10. Justify the worst case for binary search. Find the edit distance from the string "RELEVANT" to "ELEPHANT" using dynamic programming approach.
11. Distinguish between recursion and backtracking. Using Miller-Rabin primality test, check whether 53 is prime or not?
12. How does 0/1 Knapsack problem differ from fractional one? Find the minimum vertex cover in the following graph.



2081:

Long Answer Questions: [10 marks each]

13. Differentiate between dynamic programming and memorization. Compute the shortest path between every pair in the following graphs using Floyd Warshal algorithm.



14. What is the worst case of quick sort and how does randomize quick sort handle this problem? Sort the data { -2, 4, -3, 6, 12, 10, 11, 13, 9} using quick sort.
15. Does greedy algorithm guarantee optimal solution? Solve the Fractional knapsack problem to find maximum loot from given information.

Item	1	2	3	4	5	6	7
Value	12	10	20	15	2	3	50
Weight (kgs)	2	1	3	2	12	10	1

Short Answer Questions: [5 marks each]

16. Given a set A= (5,7,10,12,15,18,20}, find the subset that sum to 35 using backtracking.

17. Solve the following recurrence relations using master's method.

    a. T(n)=2T(n/2) +n3, n>1

       T(n)=1, n=1

    b. T(n)=2T(n/4) +1, n>1

       T(n)=1, n=1

18. Write an algorithm to find the nth Fibonacci number with its time and space complexity.

19. Define order statistics problem. Find the edit distance between "cat" and "car" using dynamic programming.

20. Discuss recursion and backtracking. Analyze the complexity of Miller Rabin Randomized Primality test.

21. Solve the following linear equation using Chinese Remainder Theorem.

    x = 1 MOD 3

    x = 2 MOD 5

    x = 0 MOD 7

22. Explain the approximation algorithm for vertex cover of a connected graph with an example.

23. State cook's theorem. Discuss about problem reducibility.

24. Write short notes on:

    a. Big Oh, Big Omega, Big theta

    b. Class P, Class NP and NP-Complete

2080:

Long Answer Questions: [10 marks each]

1. What is recurrence relation? How can it be solved? Show that time complexity of the recurrence relation $T(n) = 2T(n/2) + 1$ is $O(n)$ using substitution method.
2. Write down the advantages of dynamic programming over greedy strategy. Find optimal bracketing to multiply 4 matrices of order 2,3,4,2,5.
3. Discuss heapify operation with example. Write down its algorithm and analyze its time and space complexity.

Short Answer Questions: [5 marks each]

4. Define RAM model. Write down iterative algorithm for finding factorial and provide detailed analysis.
5. Write down algorithm of insertion sort and analyze its time and space complexity.
6. Write down minmax algorithm and analyze its complexity.
7. When does greedy strategy provide optimal solution? Write down job sequencing with deadlines algorithms and analyze its complexity.
8. Suppose that a message contains alphabet frequencies as given below and find Huffman codes for each alphabet:

Symbol	Frequency
a	30
b	20
c	25
d	15
e	35

9. Does backtracking give multiple solutions? Trace subset sum algorithm for the set {3,5,2,4,1} and sum=8.
10. Why extended Euclidean algorithm is used? Write down its algorithm and analyze its complexity.
11. Define NP-complete problems with examples. Give brief proof of the statement “SAT is NP-complete”.
12. Write short notes on:

    a. Aggregate Analysis

    b. Selection problems

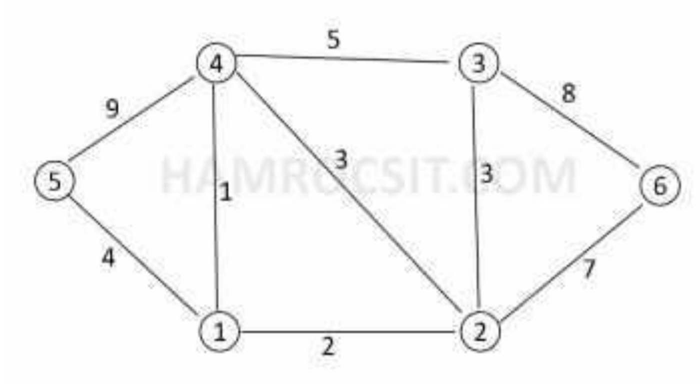
Model:

Long Answer Questions: [10 marks each]

1. Write down the elements of dynamic programming. Give the recursive definition of LCS problem. Find LCS between sequences S1 = “Dinesh”, S2 = “Dikshya”.
2. What is heap? Sort the following data items by using heap sort A [] = {3, 5, 2, 66, 4, 11, 9, 34}.
3. Given a set S = {6, 4, 5, 6, 9} and X=11. Obtain the subset sum using backtracking approach.

Short Answer Questions: [5 marks each]

4. Write down algorithm of insertion sort and analyse its time and space complexity.
5. Define binary search algorithms. Write down the recursive algorithm for binary search algorithm and analyze it.
6. Explain the asymptotic notations used to describe the time/space complexity of any algorithm.
7. What is the purpose of Euclid’s algorithm? Explain with suitable explanation.
8. Solve the recurrence relation T(n) = T(n-1) + 1 when T(0) = 0.
9. Define greedy algorithm. Find minimum spanning tree of the following graph by using Kruskal algorithm.



10. Explain in brief about the classes P, NP, and NP complete with examples.

11. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as $A_{5\times10}$, $B_{10\times15}$, $C_{15\times20}$, $D_{20\times30}$.

12. Write short notes on:

    a. Best, Worst and average case complexity

    b. Backtracking strategy

2079:

Long Answer Questions: [10 marks each]

1. Explain the divide and conquer strategy for problem solving. Describe the worst-case linear time selection algorithm and analyze its complexity.
2. Write the dynamic programming algorithm for matrix chain multiplication. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as $A_{5\times10}$, $B_{10\times15}$, $C_{15\times20}$, $D_{20\times30}$.
3. What do you mean by Backtracking? Explain the backtracking algorithm for solving 0-1 knapsack problem and find the solution for the problem given below:

Items →	1	2	3	4	
Weight (wi)→	2	3	4	5	
Profit (Pi) →	3	5	6	10	and capacity of knapsack = 10 kg

Short Answer Questions: [5 marks each]

4. Explain the iterative algorithm to find the GCD of given two numbers and analyze its complexity.
5. Generate the prefix code for the string " CYBER CRIME" using Huffman algorithm and find the total number of bits required.
6. Define tractable and intractable problems. Illustrate vertex cover problem with an example.

7. Find the edit distance between the string " ARTIFICIAL" and "NATURAL" Using dynamic programming.
8. Write short notes on:
    a. Best, Worst and average case complexity
    b. Greedy strategy
9. Solve the following recurrence relations using master's method:
    a. T(n) = 2T(n/4) + kn2, n > 1

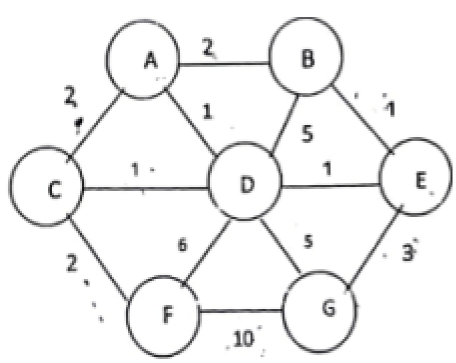
       =1, n=1
    b. T(n) = 5T(n/4) + kn, n > 1

       =1, n=1
10. Solve the following linear congruences using Chinese Remainder Theorem.

    X=1 (MOD 2)

    X=3 (MOD 5)

    X=6 (MOD 7)
11. Find the MST from following graph using Kruskal's algorithm.



12. Trace the quick sort algorithm for sorting the array A [] = {15,7,6,23, 18,34,25} and write it’s best and worst complexity.

2078:

Long Answer Questions: [10 marks each]

1. What are the elementary properties of algorithm? Explain. Why do you need algorithm? Discuss about analysis of the RAM model for analysis of algorithm with suitable example.
2. Explain about the divide and conquer paradigm for algorithm design with suitable example. Write the Quick sort algorithm using randomized approach and explain its time complexity.
3. Explain in brief about the Dynamic Programming Approach for algorithm design. How does it differ with recursion? Explain the algorithm for solving the 0/1 Knapsack problem using the dynamic programming approach and explain its complexity.

Short Answer Questions: [5 marks each]

4. Explain the recursion tree method for solving the recurrence relation. Solve following recurrence relation using this method.

   $T(n)=2T(n/2) +1$ for $n> 1$, $T(n) =1$ for $n =1$
5. Write an algorithm to find the maximum element of an array and analyze its time complexity.
6. Write the algorithm for bubble sort and explain its time complexity.
7. What do you mean by optimization problem? Explain the greedy strategy for algorithm design to solve optimization problems.

8. Explain the algorithm and its complexity for solving job sequencing with deadline problem using greedy strategy.
9. What do you mean by memorization strategy? Compare memorization with dynamic programing.
10. Explain the concept of backtracking. How does it differ with recursion?
11. Explain in brief about the complexity classes P, NP and NP Complete.
12. Write short notes on:
    a. NP Hard Problems and NP Completeness
    b. Problem Reduction

2076:

Long Answer Questions: [10 marks each]

1. What do you mean by the complexity of an algorithm? Explain the asymptotic notations used to describe the time/space complexity of any algorithm with their geometrical interpretation and example.
2. Explain the divide and conquer paradigm from algorithm design with a suitable example. Write the Quick sort algorithm using a randomized approach and explain its time complexity.
3. Explain in brief the Backtracking approach for algorithm design. How does it differ with recursion? Explain the N-Queen problem and algorithm using backtracking and analyze its time complexity.

Short Answer Questions: [5 marks each]

4. Write the algorithm for selection sort and explain its time and space complexity.
5. Solve the following recurrence relation using the master method.

    a. T(n) = 7 T(n/2) + n2
    b. T(n) = 4 T(n/4) + kn

6. Explain the greedy algorithm for the fractional knapsack problem with its time complexity.
7. Trace heap sort algorithm for the following data:

{2, 9, 3, 12, 15, 8, 11}

8. What do you mean by Dynamic programming strategy? Explain the element of DP.
9. Explain the approximation for solving vertex cover with a suitable example.
10. Explain Prism's algorithm for MST problem and analyze its time complexity.
11. Explain in brief about the classes P, NP, and NP complete with examples.
12. Write short notes on

    a. Backtracking strategy

    b. Tractable and Intractable Problem